SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:25

Code of Conduct:

I E.SHANI PRIYA (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student:

Shani priya

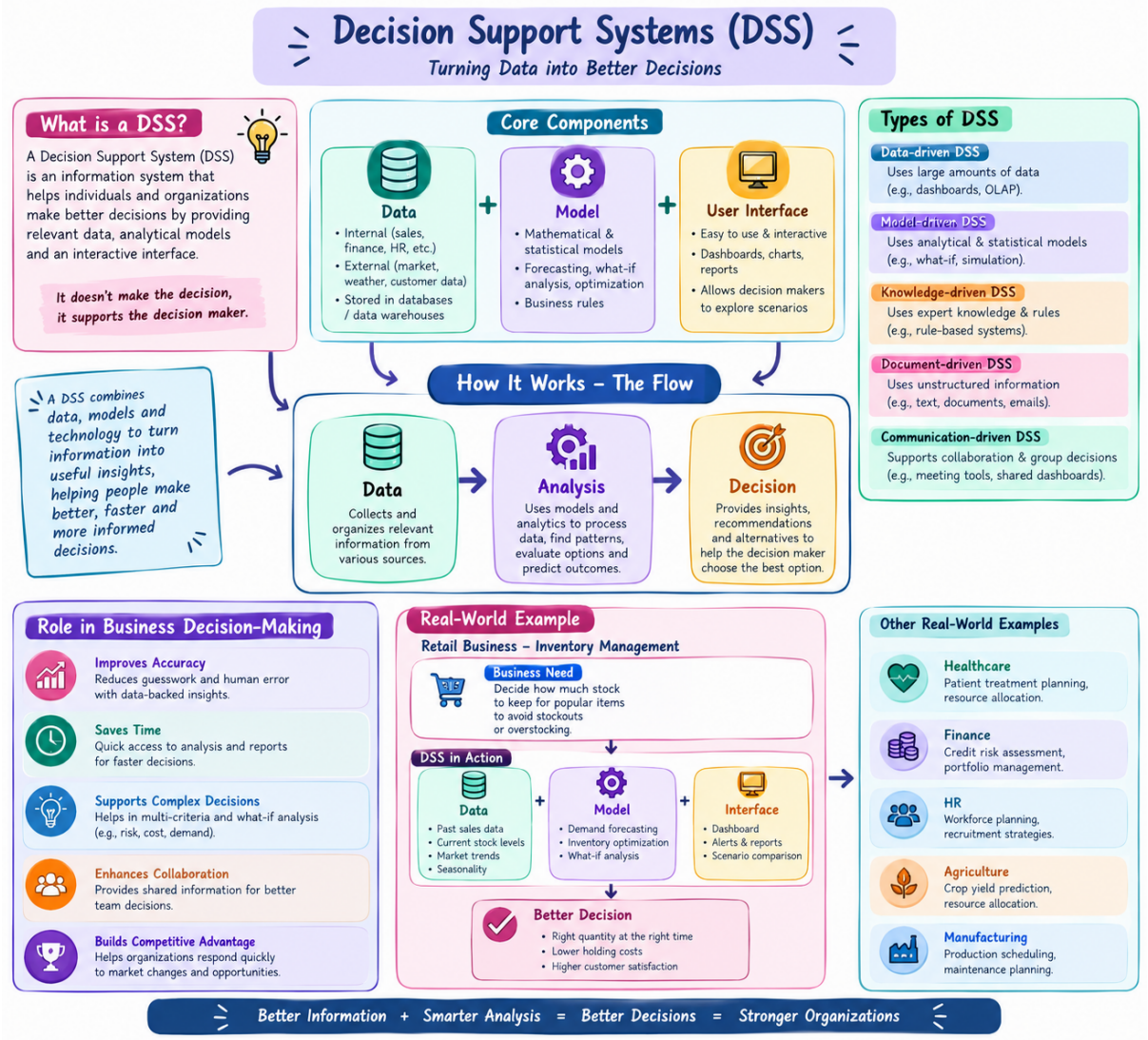
Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

What your map must show:

1. Core components: Data, Model, User Interface
2. Types of DSS
3. Flow: Data → Analysis → Decision
4. Role in business decision-making
5. Real-world example linkage



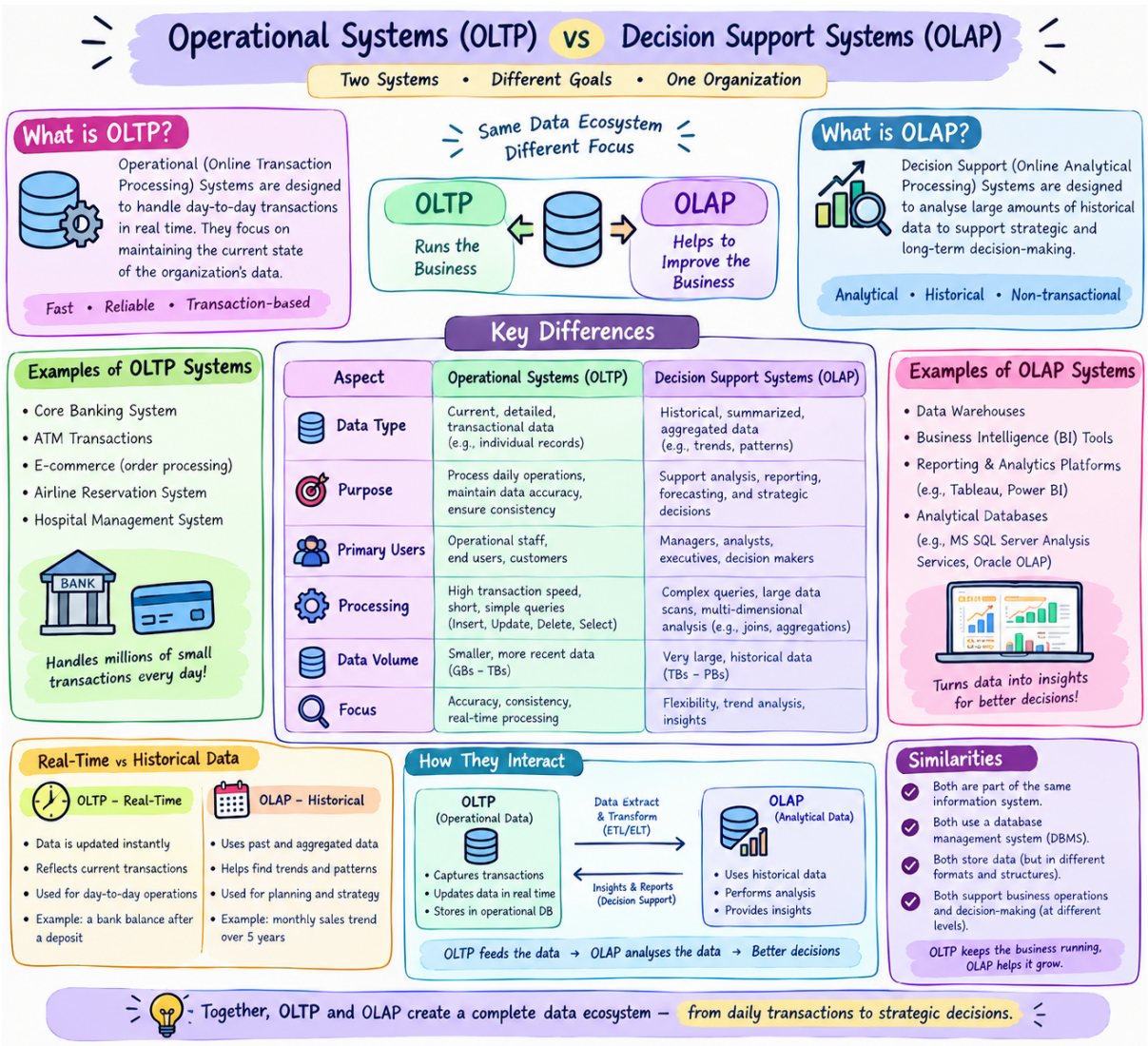
Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)



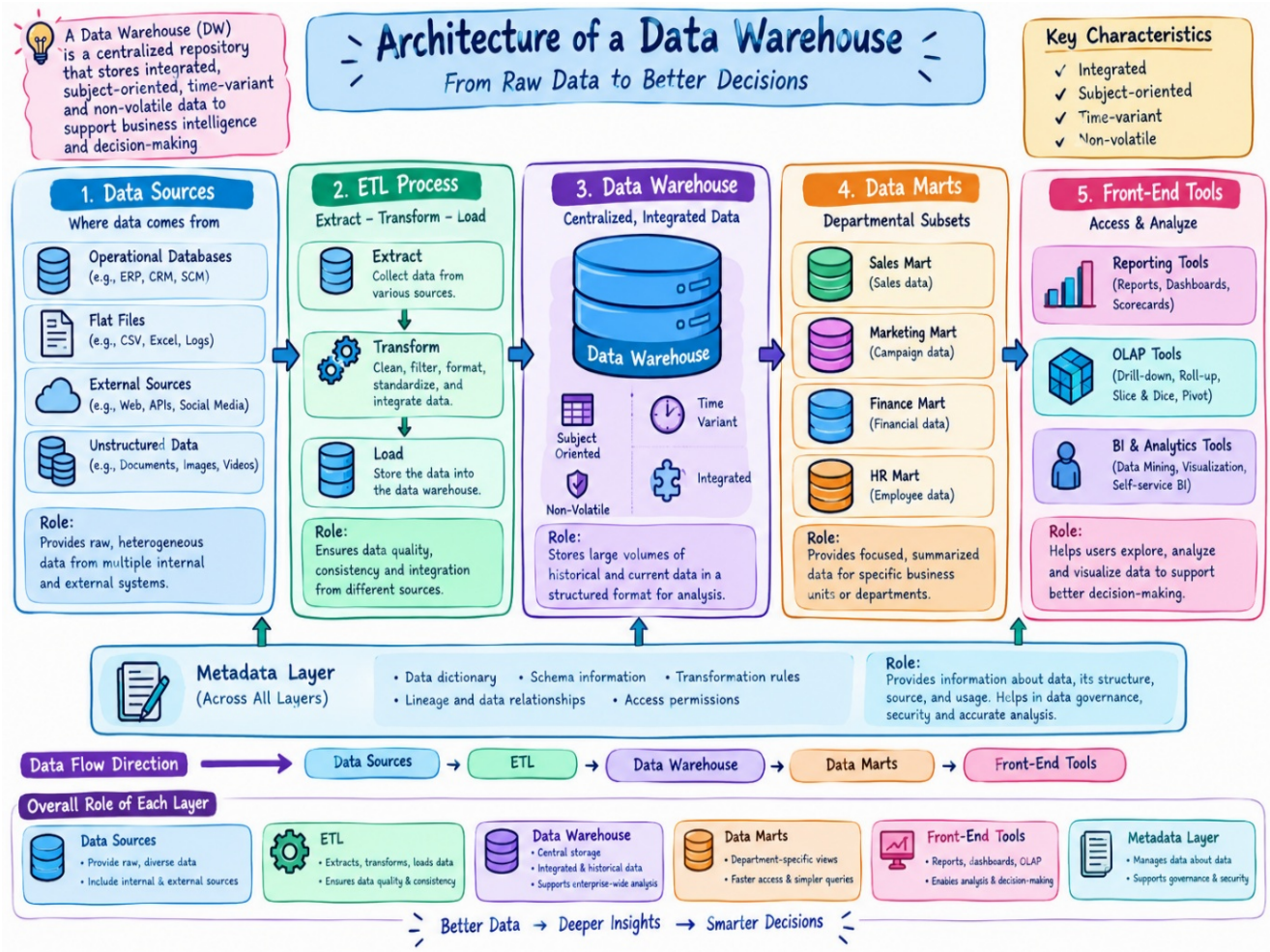
Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts
2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer



Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

What your map must show:

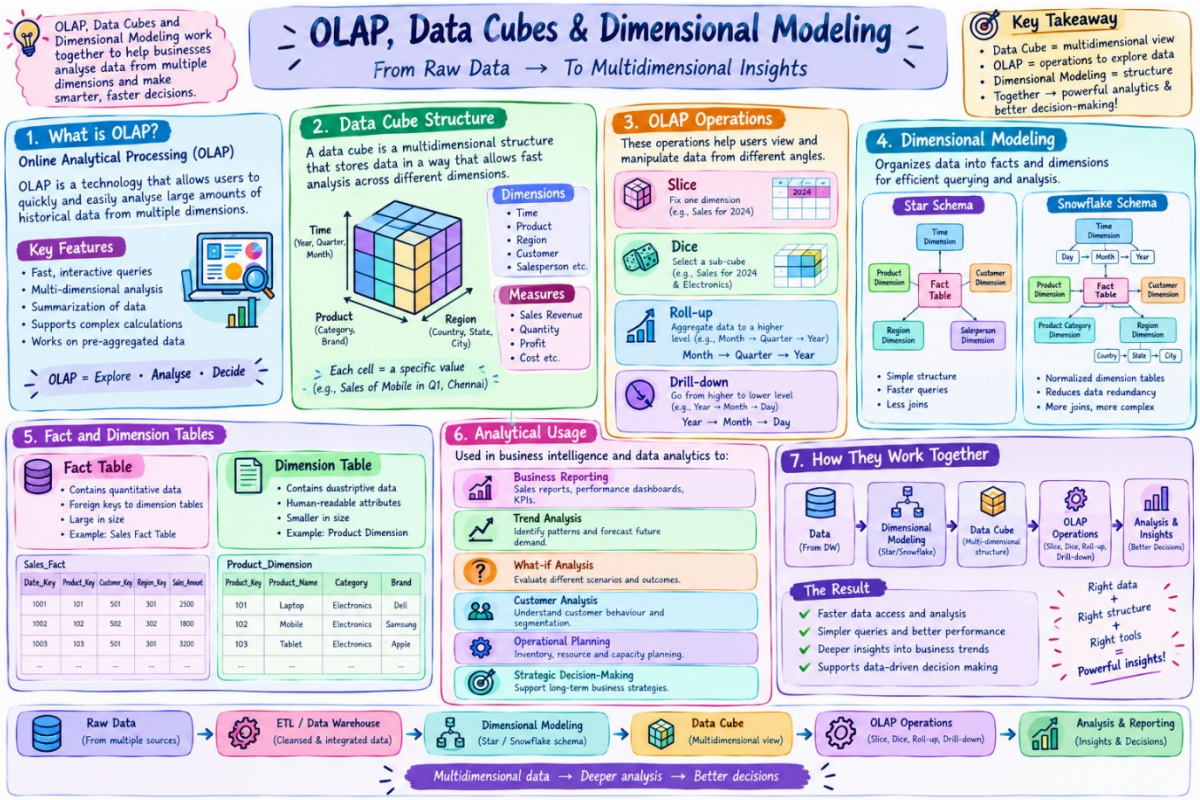
1. Data cube structure (dimensions, measures)

2. OLAP operations (slice, dice, roll-up, drill-down)

3. Star schema and snowflake schema

4. Fact and dimension tables

5. Analytical usage



Concept Map 5: Query Processing & BI Tools (KNIME)

Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

What your map must show:

1. Query processing stages (Parsing, Optimization, Execution)

2. Role of BI tools

3. KNIME workflow structure

4. Data → Query → Visualization pipeline

5. OLAP application integration.

